

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question
 $3a + 4b = 25$

A shipping company charged a customer \$25 to ship some small boxes and some large boxes. The equation above represents the relationship between a, the number of small boxes, and b, the number of large boxes, the customer had shipped. If the customer had 3 small boxes shipped, how many large boxes were shipped?

- Answer**
- A. 3
 - B. 4
 - C. 5
 - D. 6

Correct Answer: B

Rationale

Choice B is correct. It’s given that a represents the number of small boxes and b represents the number of large boxes the customer had shipped. If the customer had 3 small boxes shipped, then $a = 3$. Substituting 3 for a in the equation $3a + 4b = 25$ yields $3(3) + 4b = 25$ or $9 + 4b = 25$. Subtracting 9 from both sides of the equation yields $4b = 16$. Dividing both sides of this equation by 4 yields $b = 4$. Therefore, the customer had 4 large boxes shipped.

Choices A, C, and D are incorrect. If the number of large boxes shipped is 3, then $b = 3$. Substituting 3 for b in the given equation yields $3a + 4(3) = 25$ or $3a + 12 = 25$. Subtracting 12 from both sides of the equation and then dividing by 3 yields $a = \frac{13}{3}$. However, it’s given that the number of small boxes shipped, a, is 3, not $\frac{13}{3}$, so b cannot equal 3. Similarly, if $b = 5$ or $b = 6$, then $a = \frac{5}{3}$ or $a = \frac{1}{3}$, respectively, which is also not true.